ADITYA ABHAY SIDHAM

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EDUCATION

University of Southern California; Los Angeles, CA **GPA: 3.52/4.0**
Master of Science in Applied Data Science Expected Dec 2026

DY Patil Ramrao Adik Institute of Technology; Navi Mumbai, India **GPA: 3.68/4.0**
Bachelor's in Computer Engineering; Specialization in Data Science Jun 2024

TECHNICAL SKILLS

Languages: Python, Java, C#, TypeScript, JavaScript, C++, SQL, Swift, Bash
Backend & Frontend: FastAPI, Spring Boot, ASP.NET, REST APIs, Microservices, Celery, React, SwiftUI
Data: PostgreSQL, MySQL, SQL Server, MongoDB, Redis, Pandas, Apache Spark, ETL Pipelines
AI & Vision: Claude/Gemini APIs, Multi-agent Orchestration, RAG, MCP, MediaPipe, OpenCV, Claude Code
Cloud & DevOps: AWS (EC2, S3, Lambda, API Gateway, CloudWatch), Docker, Linux, Git, CI/CD
Engineering: Unit/Integration Testing, pytest, LLM Evaluation, Code Reviews, Design Patterns

EXPERIENCE

LTM Limited - (LTIMindtree) **Mumbai, India**
Software Engineer Sep 2024 – Dec 2024

- Built **C#/ASP.NET and REST integrations** for Otis's enterprise application, connecting service management, dispatch, inventory, and reporting so operational records could move across **four business functions**.
- Cut backend issue-resolution turnaround by **20%** by rebuilding **SQL Server** fault-event ingestion procedures, optimizing telemetry and maintenance-history joins, and correcting status updates for downstream work orders.

Tata Consultancy Services **Navi Mumbai, India**
Software Development Intern Jan 2024 – May 2024

- Reduced manual data-entry time by **35%** for the Central Plan Scheme Monitoring project by building a **Java/Spring Boot** module with SQL automation and form validation in **two-week Agile sprints**.

Aces Invensys Technologies **Navi Mumbai, India**
IoT Intern May 2023 – Jun 2023

- Built a **Python/MQTT** pipeline to collect live beacon signals and calculate positions, achieving **approximately ±10 cm accuracy in testing** through sensor communication protocols and positioning analytics.

PROJECTS

FitandFine: AI Nutrition Platform | *SwiftUI, FastAPI, PostgreSQL, Redis, Celery* **Apr 2026 – Present**

- Solo-built an iOS app with **sub-500 ms barcode lookup** across **50,000+ USDA records** and a **four-agent Claude workflow** exceeding **85% label-field accuracy**; offloaded model calls to Celery workers to keep requests responsive.
- Automated catalog **ETL with S3, Lambda, and API Gateway**, cutting refreshes from **25 to 8 minutes**; added retry logic, **CloudWatch alarms**, and unit/integration tests with **85% backend line coverage**.

Chat-DB: Conversational Database Explorer | *Python, Gemini, MySQL, MongoDB* **Mar – May 2025**

- Led a **four-person team** building a Streamlit app that cut query-construction time by **approximately 60%** for non-SQL users; combined **live schema discovery** with separate prompts for SQL and MongoDB aggregation pipelines.
- Built editable query previews and confirmation before execution; added a **pytest evaluation suite of 150 labeled questions**, reaching **92% result correctness** and catching **12 regressions** before release.

FitForm: Real-Time Exercise Coach | *Python, MediaPipe/BlazePose, OpenCV* **2024**

- Turned an ordinary webcam into a form-aware coach using **33 body landmarks**, joint-angle geometry, and **exercise-specific state machines**; counted full-range reps across **3+ exercises** with live visual corrections on CPU.
- Added confidence filtering and temporal smoothing to sustain **25 FPS on a laptop CPU** and cut false rep counts by **30% across 120 annotated clips**, improving feedback consistency during fast or partially occluded movements.

PATENT & CERTIFICATION

- **FitForm:** Co-applicant, published Indian patent application **202421099595** (Jan 2025), for workout-form correction using computer vision. **Google Data Analytics Professional Certificate** (Jun 2024).